The Schematics have been analyzed from
a die picture by John McMaster.
The chip is identified as:

Fujitsu
672172
C 1985

It is a C1200AVB, MB672172 device
from the AVB-CMOS series.
- 1245 Basic Cells
- 15 Columns, 83 Rows

Cells used (more complex ones):

FJ4 - Power J-K FF with Clear

C41 - 4-bit Binary Asynchronous Counter

FJ5 - Power J-K FF with Clear and Preset

C43 - 4-bit Binary Synchronous Up Counter

FDO - DFF with Reset

FDN - DFF with Set

V1N - Inverter Buffer

K2B - Power Clock Buffer

BD3 - Buffer (Delay Cell)

Delay 11.8ns

V2B - Power inverter

O4T - Output Buffer (Hi-Z)

Not sure this is
the correct type.

Clocks and Reset

File: clocks_and_reset.kicad_sch

Horizontal Signals

File: horizontal_signals.kicad_sch

Vertical Signals

File: vertical_signals.kicad_sch

DMA

File: dma.kicad_sch

DB0 is connected externally to D0 of the data bus

Sync and Blanking

File: sync_and_blanking.kicad_sch

Data Acknowledge

File: data_acknowledge.kicad_sch

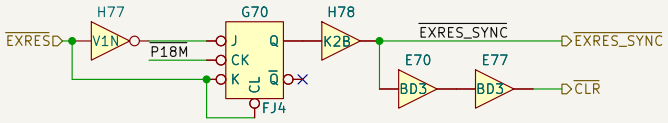
Ulf Skutnabba, twitter: @skutis77

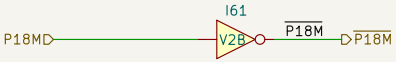
Sheet: /
File: 007782.kicad_sch

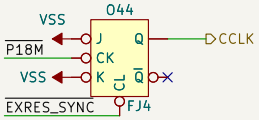
Title: Konami 007782

Size: A4	Date: 2023-08-09	Rev:
KiCad E.D.A. 10.0.4		Id: 1/7

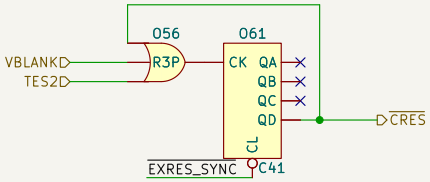
External reset synchronized to P18M clock.







The frequency of CCLK is the
frequency of P18M divided by 2.
f_CCLK = f_P18M / 2 = 18.432Mhz / 2 = 9.216MHz.

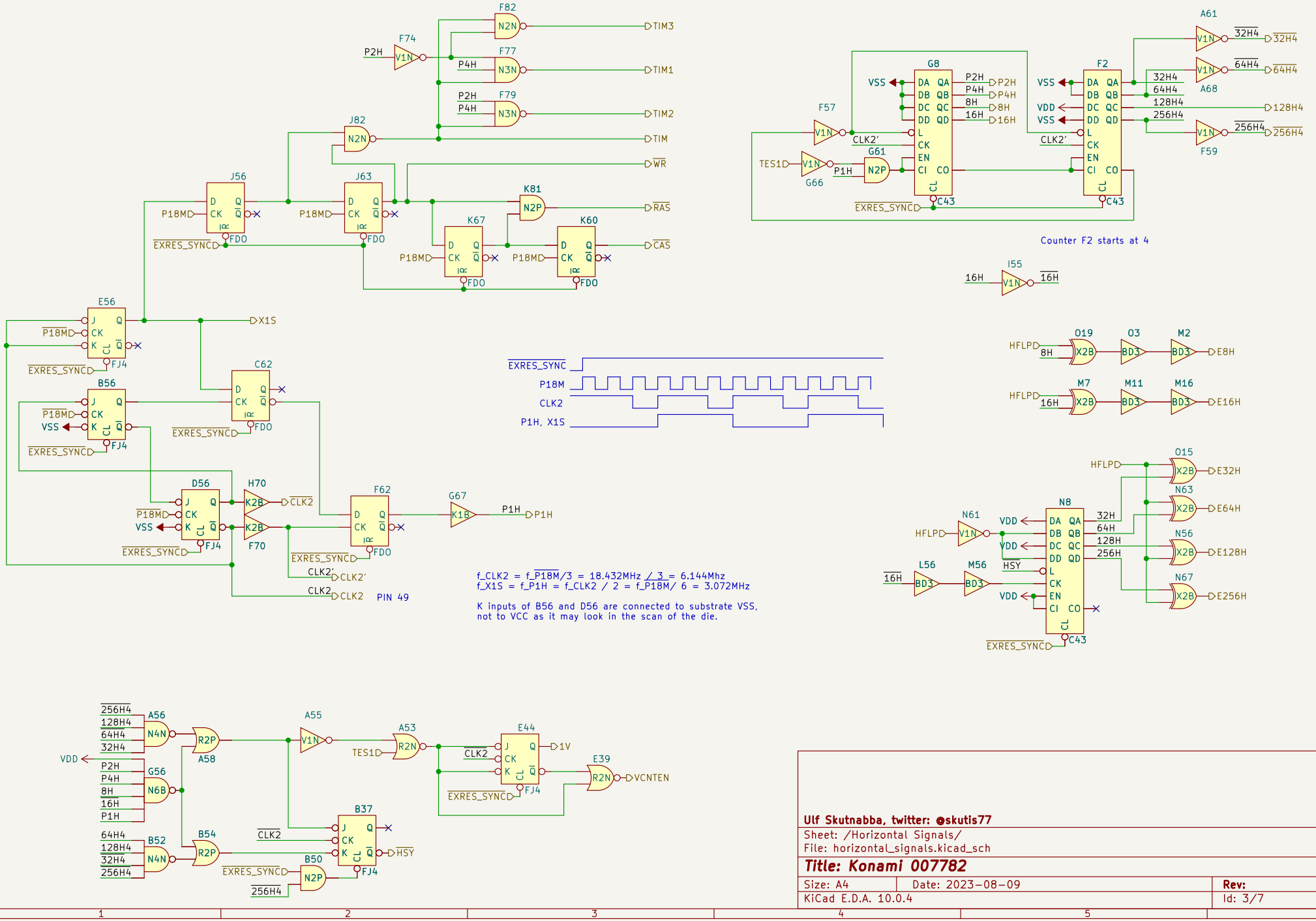


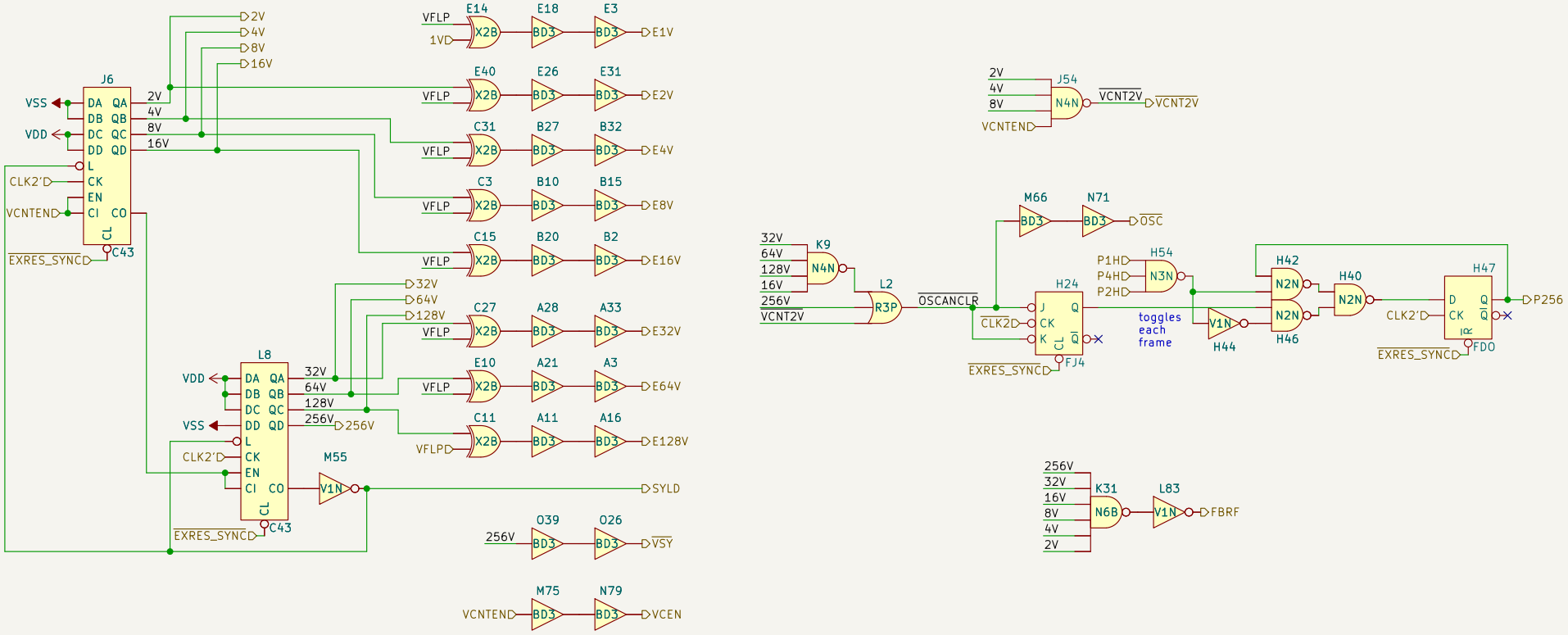
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Sheet: /Clocks and Reset/
File: clocks_and_reset.kicad_sch

Title: Konami 007782

Size: A4	Date: 2023-08-09	**Rev:**
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F82
N2N
TIM3
F74
P2H
V1N
F77
P4H
N3N
TIM1
F79
P2H
P4H
N3N
TIM2
J82
N2N
TIM
WR
J56
J63
K81
N2P
RAS
K67
K60
CAS
P18MD
EXRES_SYNC
FDO
G8
F2
A61
A68
F59
F57
G61
G66
TES1
P1H
CLK2'
C43
Counter F2 starts at 4
I55
E56
X1S
FJ4
C62
B56
VSS
D56
H70
F70
F62
G67
K1B
P1H
CLK2
PIN 49
EXRES_SYNC
P18M
CLK2
P1H, X1S
f_CLK2 = f_P18M/3 = 18.432MHz / 3 = 6.144Mhz
f_X1S = f_P1H = f_CLK2 / 2 = f_P18M/ 6 = 3.072MHz
K inputs of B56 and D56 are connected to substrate VSS,
not to VCC as it may look in the scan of the die.
O19
O3
M2
M7
M11
M16
HFLPD
X2B
BD3
E8H
E16H
O15
N63
N56
N67
E32H
E64H
E128H
E256H
N8
N61
L56
M56
HSY
A56
A58
A55
A53
E44
E39
VCNTEN
G56
B52
B54
B37
B50
VDD
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Sheet: /Horizontal Signals/
File: horizontal_signals.kicad_sch
Title: Konami 007782
Size: A4
Date: 2023-08-09
Rev:
KiCad E.D.A. 10.0.4
Id: 3/7

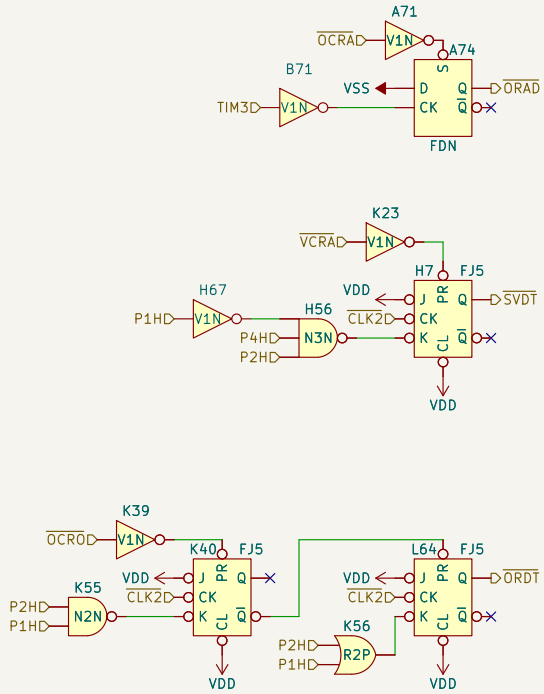


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Sheet: /Vertical Signals/
File: vertical_signals.kicad_sch

Title: Konami 007782

Size: A4	Date: 2023-08-09	**Rev:**
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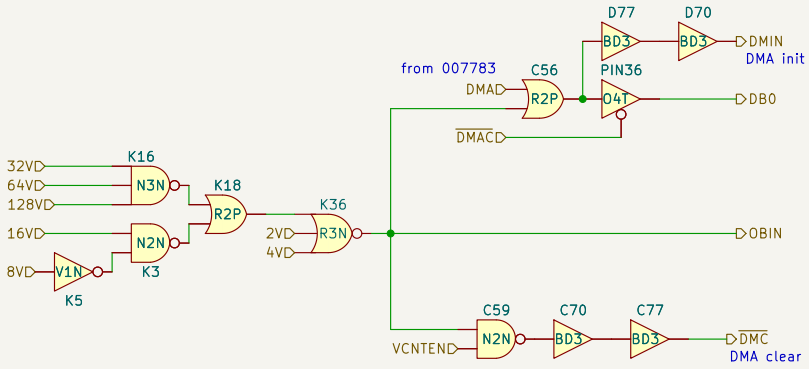


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Sheet: /Data Acknowledge/
File: data_acknowledge.kicad_sch

Title: Konami 007782

Size: A4	Date: 2023-08-09	**Rev:**
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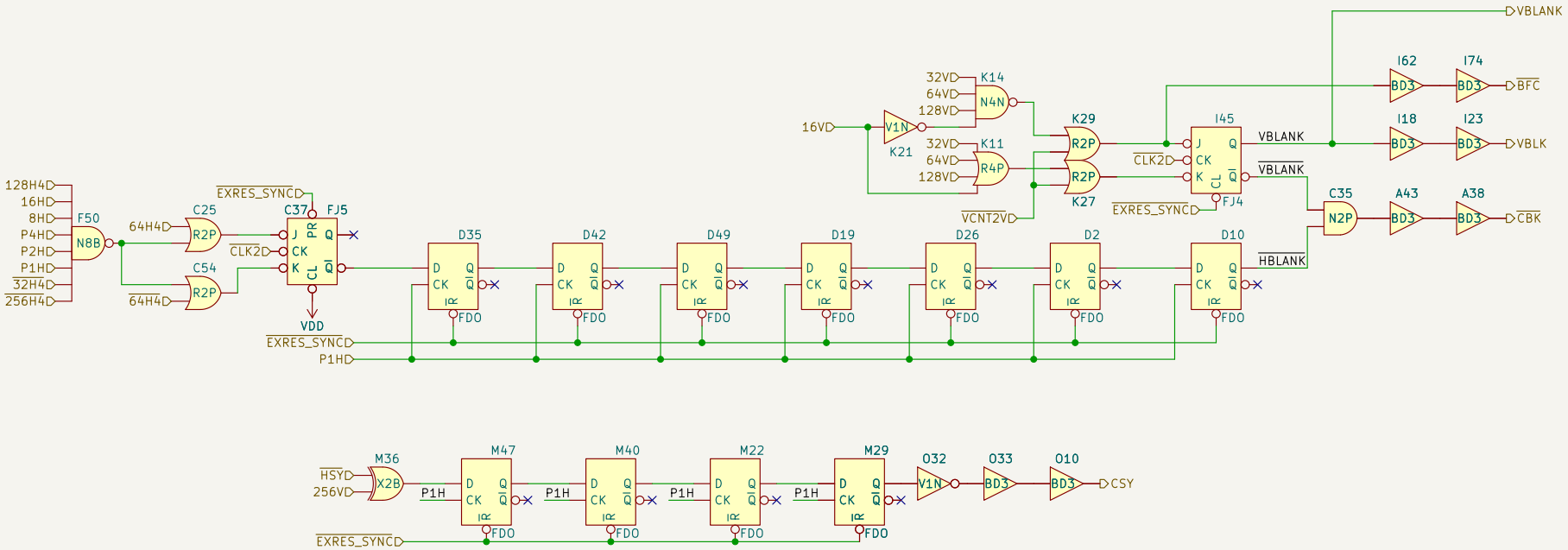


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Sheet: /DMA/
File: dma.kicad_sch

Title: Konami 007782

Size: A4	Date: 2023-08-09	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/7



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Sheet: /Sync and Blanking/
File: sync_and_blanking.kicad_sch

Title: Konami 007782

Size: A4	Date: 2023-08-09	**Rev:**
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